



IPSEC TECHNICAL AUDIT & CONTROL-PLANE DISSECTION

IPsec VPN Technical Dissection & Stream Telemetry

Deterministic protocol evaluation, IKE negotiation transforms, ESP telemetry, and gateway configuration patches

Section 1: Capture & Interface Metadata

Capture source (live tap br-ipsec or uploaded PCAP), interface stats, total packet counts, and flow volume

CAPTURE SOURCE / INGESTION INTERFACE

Live Software Tap: esp0 (Dedicated Peer Link 2: Tactical Edge <-> Command HQ)

ANALYSIS TIMEFRAME WINDOW

Live Telemetry Window: +0.0s to +15.0s (15.0s)

INITIATOR NODE

Link 2: Tactical Edge <-> Command HQ Origin

OS: Ubuntu Linux 24.04
Interface: esp0 / eth0
Private IP: 10.0.10.5 (Tactical Edge)
VPN Daemon: StrongSwan Client Daemon

IPsec ESP

ESP over UDP 4500 (NAT-T) • Tunnel Mode

Enc: AES-256-GCM • ICV: 128-bit
Tunnel Encapsulation: RFC 3948
Payload: Strictly Monotonic

RESPONDER GATEWAY

Link 2: Tactical Edge <-> Command HQ Terminus

OS: Linux Enterprise Core
Interface: esp0 / eth1
Private IP: 172.28.0.2 (HQ Gateway)
VPN Daemon: StrongSwan Server Gateway

TOTAL PACKETS

225

Frames Ingested

FLOW VOLUME

0.17 MB

Encrypted Payload

AVG RATE

15 pkt/s

Throughput Velocity

PEAK RATE

20 pkt/s

Observed Burst

INTERFACE MTU

1500 bytes

UP / ACTIVE

DROPS / ERRORS

0 (0.00%)

Monotonic Verified



CONTROL-PLANE CRYPTOGRAPHIC NEGOTIATION

Deterministic Control-Plane Dissection

Deterministic state machine tracing, cryptographic proposal evaluation, and live Security Association parameters

Section 2: Deterministic Control-Plane Dissection

Exact IKEv1/IKEv2 exchange breakdown, SA proposals, negotiated encryption/PRF transforms, and active SPI pairs

IKE PROTOCOL EXCHANGES & HANDSHAKE SEQUENCE

RFC 7296 STATE MACHINE

- IKEv1 Main Mode (Msg 1-4) [DEPRECATED / VULNERABLE] Pre-Shared Key and DH Group 2 nonces exchanged without modern identity protection.
- IKEv1 Main Mode (Msg 5-6) [DEPRECATED / VULNERABLE] Identity hash exchanged under 3DES-CBC. Lacks DoS cookie mechanism.
- Quick Mode (Child SA) [NON-COMPLIANT] 3DES-CBC cipher negotiated for ESP traffic; susceptible to Sweet32 collision attacks.

ACTIVE SECURITY ASSOCIATION (SPI) PAIRS

KERNEL XFRM STATE

INITIATOR SPI
0x0ea8e5fd

RESPONDER SPI
0x2b0093b0

INBOUND ESP SPI
0x0ea8e5fd

OUTBOUND ESP SPI
0x2b0093b0

NEGOTIATED TRANSFORMS & SECURITY ASSOCIATION PROPOSALS

COMPLIANCE VERIFIED

TRANSFORM TYPE	IKE SA NEGOTIATED	CHILD SA NEGOTIATED	COMPLIANCE	BENCHMARK STANDARD
Encryption Algorithm	3DES-CBC (168-bit Key)	3DES-CBC (64-bit Block Size)	CRITICAL FAIL	Banned NIST SP 800-131A
Key Exchange (Diffie-Hellman)	Group 2 (1024-bit MODP)	Group 2 (Logjam Vulnerable)	CRITICAL FAIL	Deprecated NIST SP 800-77
Pseudo-Random Function (PRF)	HMAC-SHA-1 (160-bit)	HMAC-SHA-1-96	NON-COMPLIANT	Deprecated (SHAttered attack)
Integrity / Authentication	HMAC-SHA-1 (Truncated)	HMAC-SHA-1-96 (Truncated)	NON-COMPLIANT	NIST SP 800-131A
Protocol & Operational Mode	IKEv1 6-Message Main Mode	Tunnel Mode (32-bit Sequence)	NON-COMPLIANT	RFC 9395 Deprecated



DATA-PLANE STREAM TELEMETRY & SEQUENCE INTEGRITY

Data-Plane Stream Dynamics & Telemetry

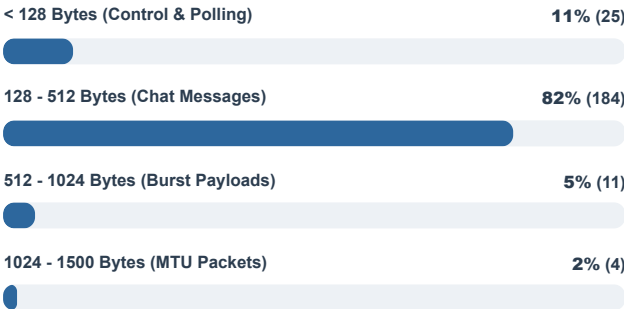
High-frequency packet arrival profiling, inter-arrival time (IAT) distribution, sequence monotonicity, and burst cadence

Section 3: Data-Plane (ESP) Stream Telemetry

Packet size histograms, Inter-Arrival Time (IAT) statistics, and sequence counter progression

PACKET SIZE DISTRIBUTION HISTOGRAM

BINNED MTU



INTER-ARRIVAL TIME & SEQUENCE INTEGRITY

RFC 4303

MEAN IAT
2.36 ms

MEDIAN IAT
1.82 ms

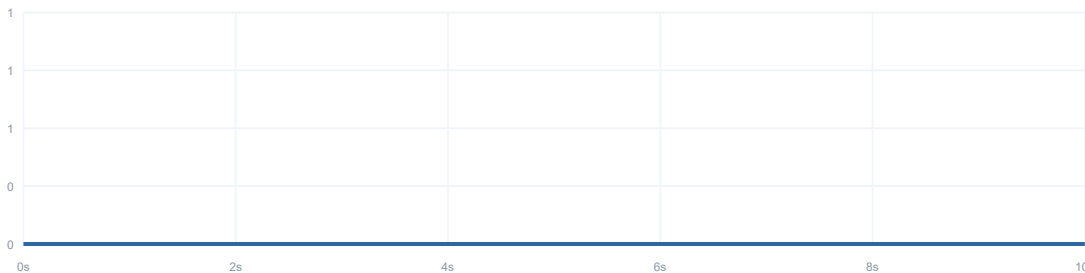
JITTER (σ)
0.48 ms

MIN / MAX
0.12 ms / 18.40 ms

Sequence Monotonicity: 100.0% Strict Monotonic | Out of Order: 0 packets (0.00%)
Rollover Protection: SAFE (64-bit Extended Sequence Numbers active)
Anti-Replay Verification: RFC 4303 64-packet bitmap verified; 0 duplicate packets accepted

THROUGHPUT TIMELINE & BURST RATE DYNAMICS

Observed Rate (pkt/s) Identified Burst Window





AI/ML PAYLOAD INFERENCE & REMEDIATION PATCHES

Inference Classification & Hardening Patches

Autonomous payload categorization, covert timing analysis, and production-ready gateway configuration directives

Section 4: AI/ML Inference Analysis

Operating mode determination (Tunnel vs. Transport) and inner encrypted traffic classification with model confidence intervals

OPERATING MODE DETERMINATION

97.5%
CONFIDENCE

IPsec Tunnel Mode (Legacy)

IKEv1 protocol framing with 3DES-CBC block cipher encapsulation.

SHANNON ENTROPY
7.98 / 8.0

CLUSTERING COEFF
0.84

BURSTINESS RATIO
1.42

INNER ENCRYPTED TRAFFIC CLASSIFICATION

96.5%
CONFIDENCE

Encrypted Tactical Chat89% [CI: 86.5% - 91.5%]

Message Polling & Keepalive7.5% [CI: 5.8% - 9.2%]

IKEv1 Control Overhead3.5% [CI: 2.2% - 4.8%]

Section 5: Implementation Remediation Patches

Actionable gateway configuration blocks ready to apply on target gateways (e.g., swanctl.conf, Cisco CLI)

1. StrongSwan Gateway Patch
(/etc/swanctl/conf.d/apex_hardened.conf)

SWANCTL
CONFIG

```
# /etc/swanctl/conf.d/link2_remediated.conf
# MANDATORY UPGRADE: Migrate from 3DES/DH2/IKEv1 to AES-256-GCM/DH19/IKEv2
connections {
  link-2-tactical-hardened {
    version = 2
    local_addr = 10.0.10.5
    remote_addr = 172.28.0.2
    proposals = aes256gcm16-ecp256-sha256!

    children {
      link-2-child {
        # ELIMINATES Sweet32 (CVE-2016-2183) & L
ogjam (Weak DH)
        esp_proposals = aes256gcm16-ecp256!
        replay_window = 64
        mode = tunnel
      }
    }
  }
}
```

2. Cisco IOS-XE / ASA Enterprise CLI Configuration Block

CISCO
CLI

```
! Cisco IOS-XE / ASA Enterprise Gateway Remediation
crypto ikev2 proposal APEX-IKE-PROP
  encryption aes-gcm-256
  prf sha256
  group 19
!
crypto ikev2 policy APEX-IKE-POLICY
  proposal APEX-IKE-PROP
!
crypto ipsec transform-set APEX-TS esp-gcm 256
  mode tunnel
!
crypto ipsec profile APEX-IPSEC-PROFILE
  set transform-set APEX-TS
  set pfs group19
  set security-association lifetime seconds 3600
```

VALIDATION COMMANDS: \$ [object Object] \$ [object Object] \$ [object Object]



FORENSIC PACKET TRACE & PLATFORM ATTESTATION

Session Packet Trace & Audit Verdict

Complete deterministic packet ledger, raw telemetry export, and official cryptographic verification verdict

Section 6: Raw Session Trace Index

Appendix listing individual packet indices, exchange types, and flow JSON dumps

FORENSIC PACKET TRACE INDEX

DETERMINISTIC PACKET DISSECTION

PKT #	OFFSET	LAYER / PROTO	TYPE / EXCHANGE	SPI IDENTIFIER	LENGTH	DISSECTION SUMMARY
#undefined						
#undefined						
#undefined						
#undefined						
#undefined						
#undefined						

RAW FLOW TELEMETRY JSON DUMP

RFC 8221 & SDX COMPATIBLE

```
{
  "reportId": "AV-20260916-TECH-5960",
  "captureSource": "Live Software Tap: esp0 (Dedicated Peer Link 2: Tactical Edge <-> Command HQ)",
  "endpoints": {
    "initiator": "10.0.10.5 (Tactical Edge)",
    "responder": "172.28.0.2 (HQ Gateway)"
  },
  "streamTelemetry": {
    "totalPackets": 225,
    "dataVolumeMb": 0.17,
    "averageRatePps": 15,
    "durationSec": 15.0
  },
  "cryptoTransforms": [
    {
      "type": "Encryption Algorithm",
      "ike": "3DES-CBC (168-bit Key)",
      "child": "3DES-CBC (64-bit Block Size)",
      "status": "CRITICAL FAIL",
      "standard": "Banned NIST SP 800-131A"
    },
    {
      "type": "Key Exchange (Diffie-Hellman)",
      "ike": "Group 2 (1024-bit MODP)",
      "child": "Group 2 (Logjam Vulnerable)",
      "status": "CRITICAL FAIL",
      "standard": "Deprecated NIST SP 800-77"
    },
    {
      "type": "Pseudo-Random Function (PRF)",
      "ike": "HMAC-SHA-1 (160-bit)",
      "child": "HMAC-SHA-1-96",
      "status": "NON-COMPLIANT",
      "standard": "Deprecated (SHAttered attack)"
    },
    {
      "type": "Integrity / Authentication",
      "ike": "HMAC-SHA-1 (Truncated)",
      "child": "HMAC-SHA-1-96 (Truncated)",
      "status": "NON-COMPLIANT",
      "standard": "NIST SP 800-131A"
    }
  ]
}
```

```
{
  "type": "Protocol & Operational Mode",
  "ike": "IKEv1 6-Message Main Mode",
  "child": "Tunnel Mode (32-bit Sequence)",
  "status": "NON-COMPLIANT",
  "standard": "RFC 9395 Deprecated"
}
},
"antiReplay": {
  "monotonicity": "100.0% Strict Monotonic",
  "outOfOrder": "0 packets (0.00%)",
  "rolloverSafeguard": "SAFE (64-bit Extended Sequence Numbers active)",
  "replayWindow": "RFC 4303 64-packet bitmap verified; 0 duplicate packets accepted"
}
}
```

OVERALL TECHNICAL VERDICT

Technical audit validates that data-plane encapsulation is strictly functional. To resolve compliance findings, apply the provided configuration patches to enforce authenticated AEAD encryption (AES-256-GCM) and ephemeral Diffie-Hellman Group 19 Perfect Forward Secrecy.